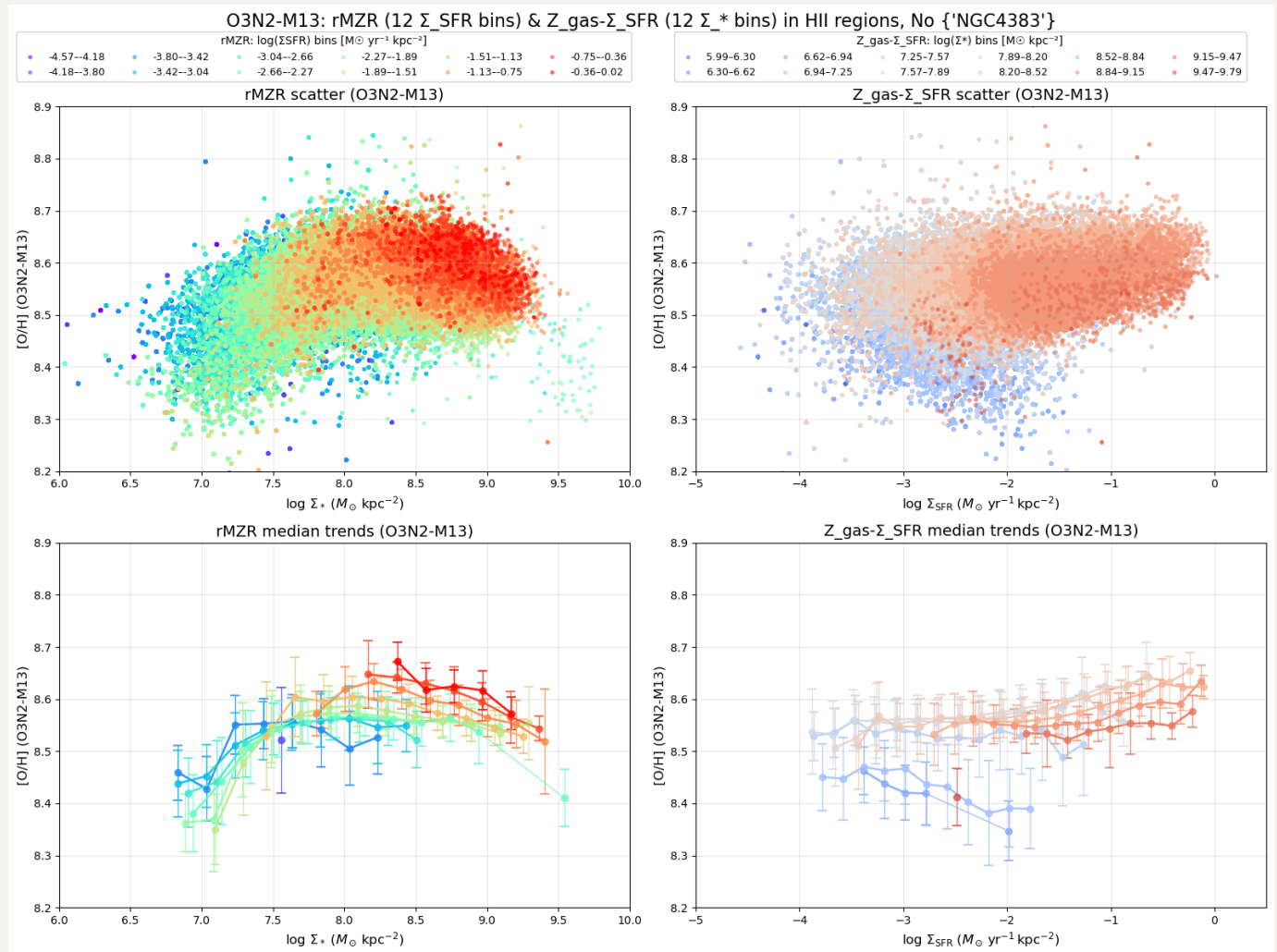


# 20251014 Offset Map from Bin Average in HII.md

Update: now I take the average from each  $\Sigma_*$  bin as the standard.

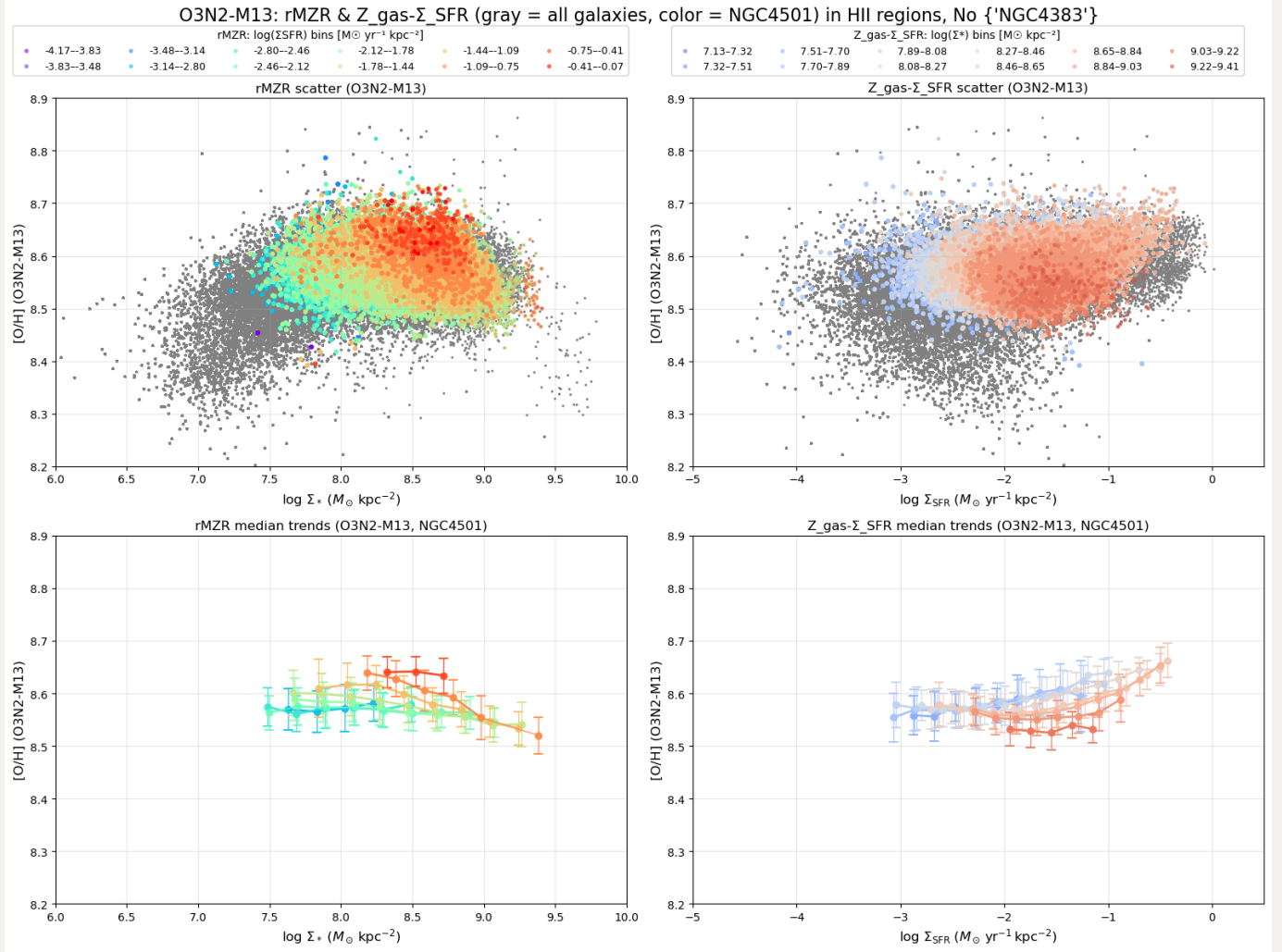
## Combined rMZR and $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation in HII regions without NGC4383

Now I update to have 12 bins.



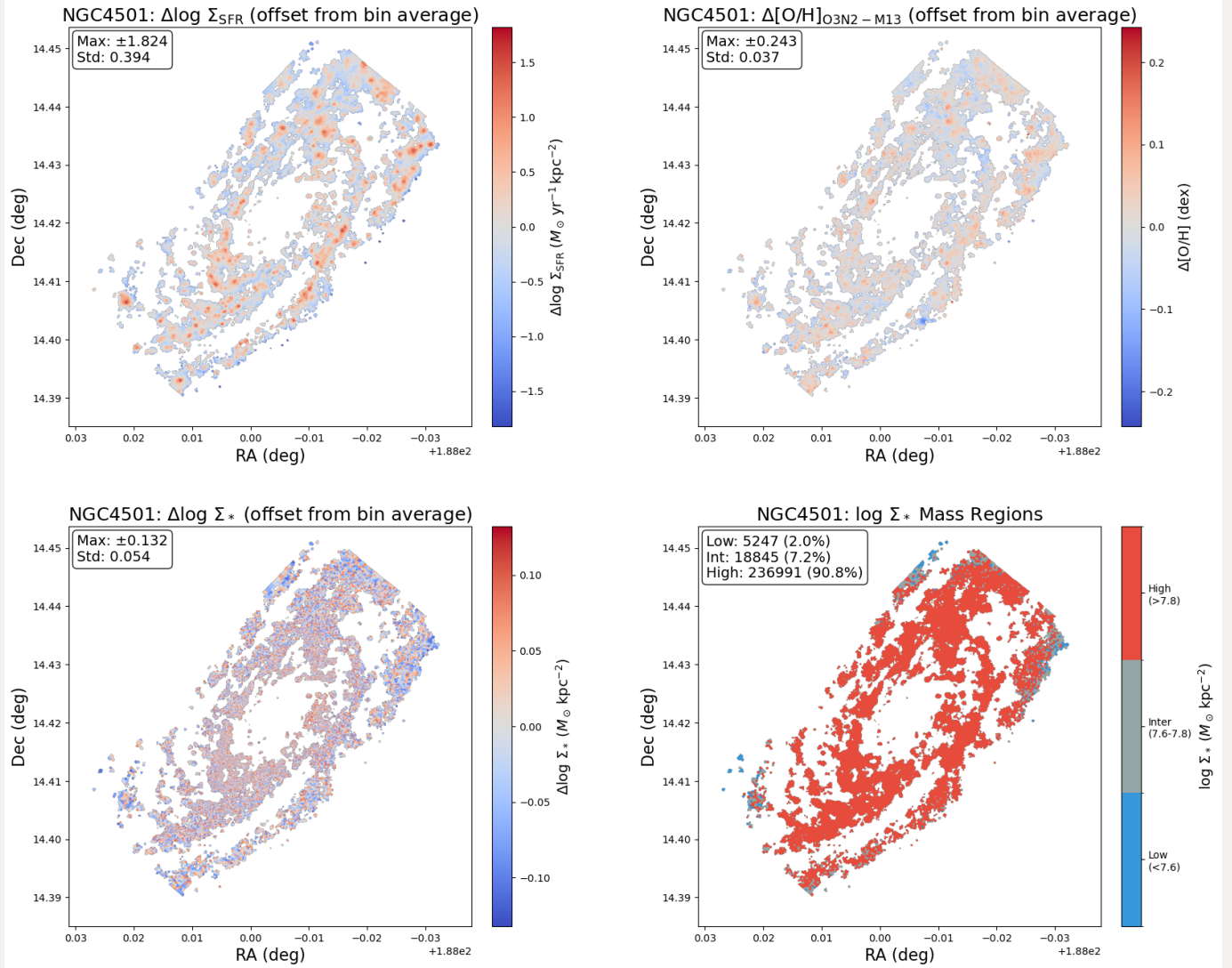
NGC4501

NGC4501 shows a clear positive correlation between  $Z_{\text{gas}}$  and  $\Sigma_{\text{SFR}}$ .



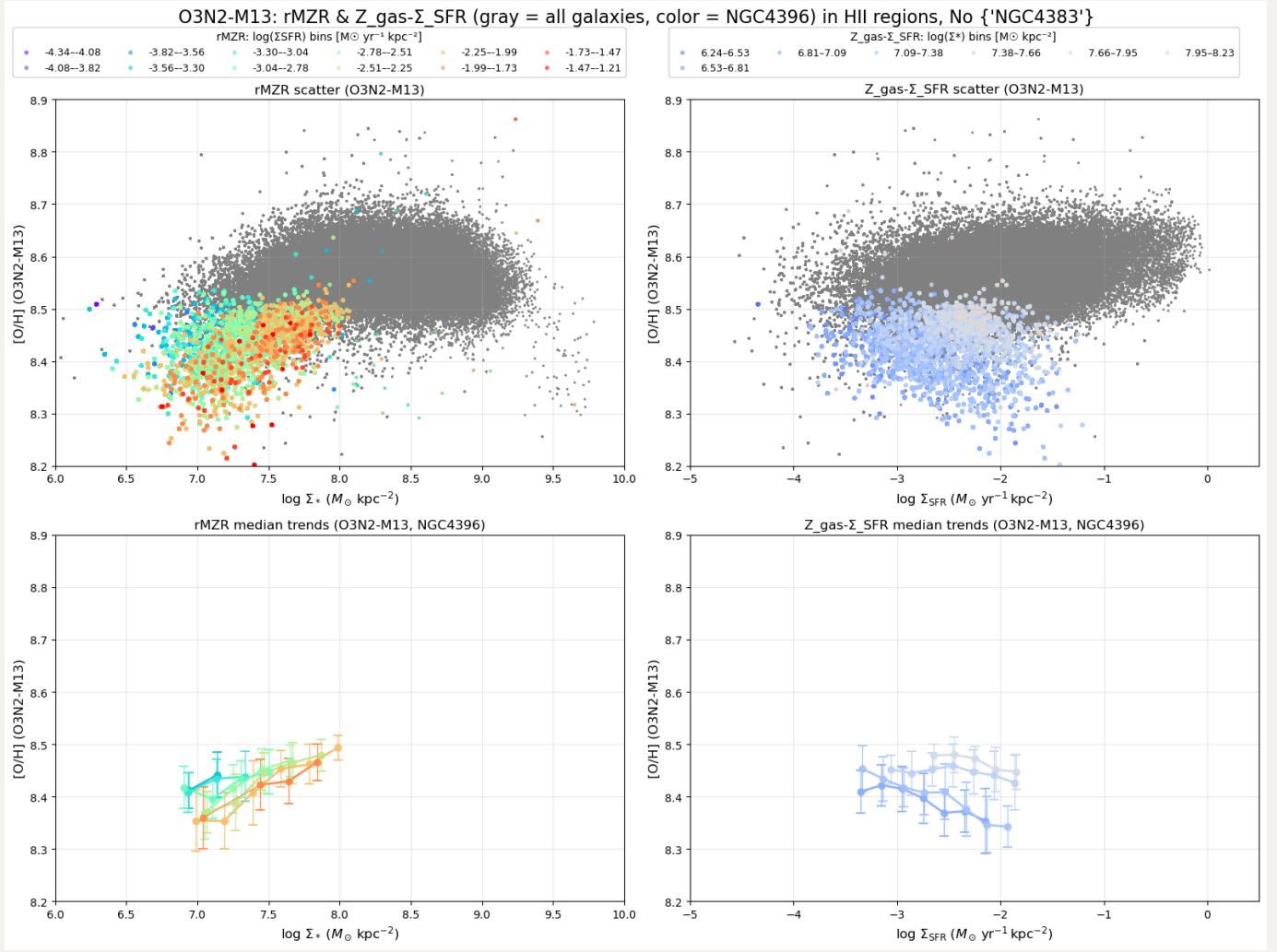
Here I try taking the median value as the standard (offset 0) and then perform the offset map to see how the  $\Sigma_{\text{SFR}}$  and  $Z_{\text{gas}}$  vary across the galaxy. For NGC4501, we can see in general, redder on the LHS also shows redder on the RHS. However, we also notice the spatial variation: the inverse correlation in this relatively low-mass HII regions.

## NGC4501: Bin-Based Offset Maps with Mass Regions

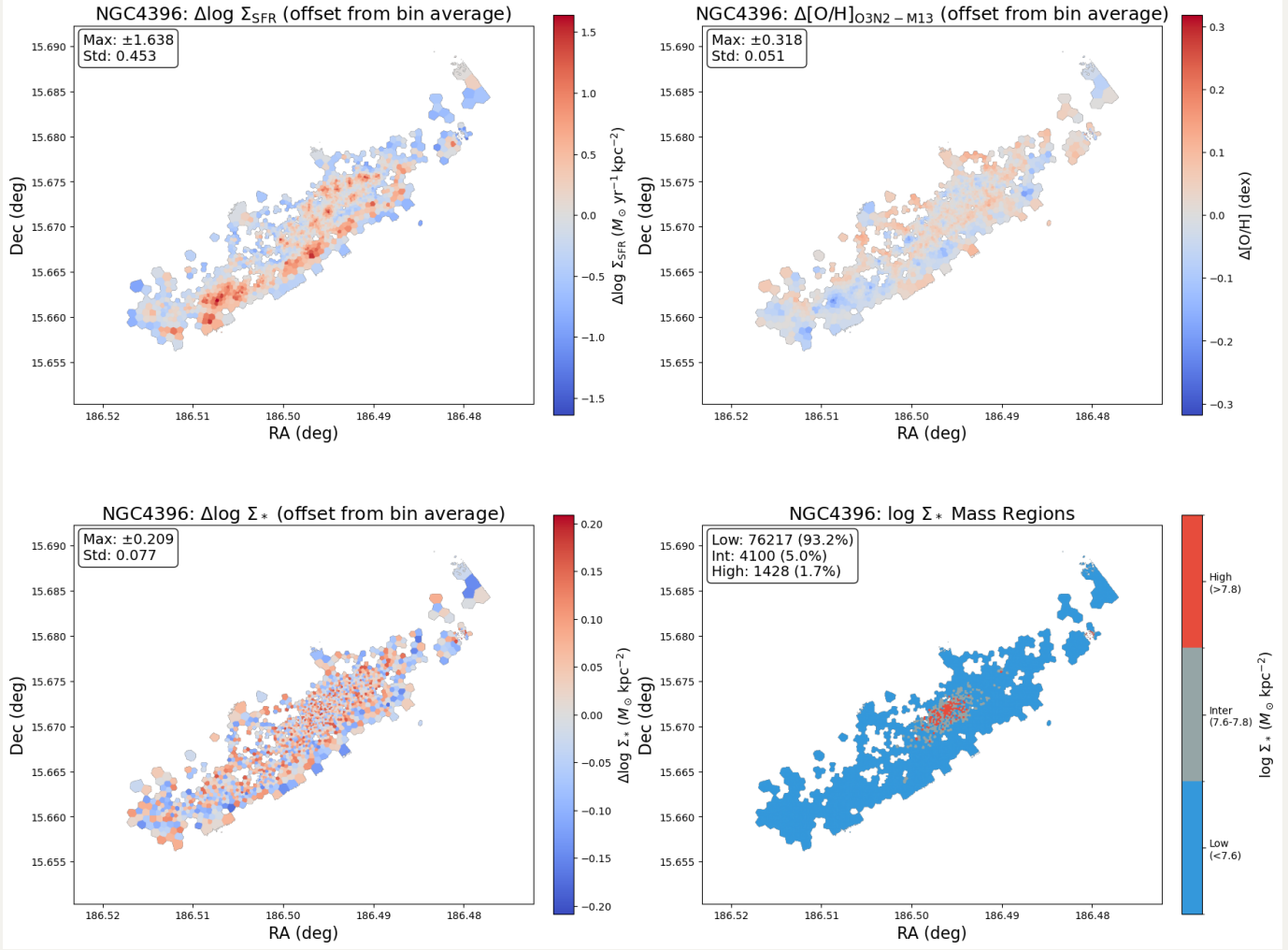


## NGC4396

NGC4396 shows a clear negative correlation between  $Z_{\text{gas}}$  and  $\Sigma_{\text{SFR}}$ .

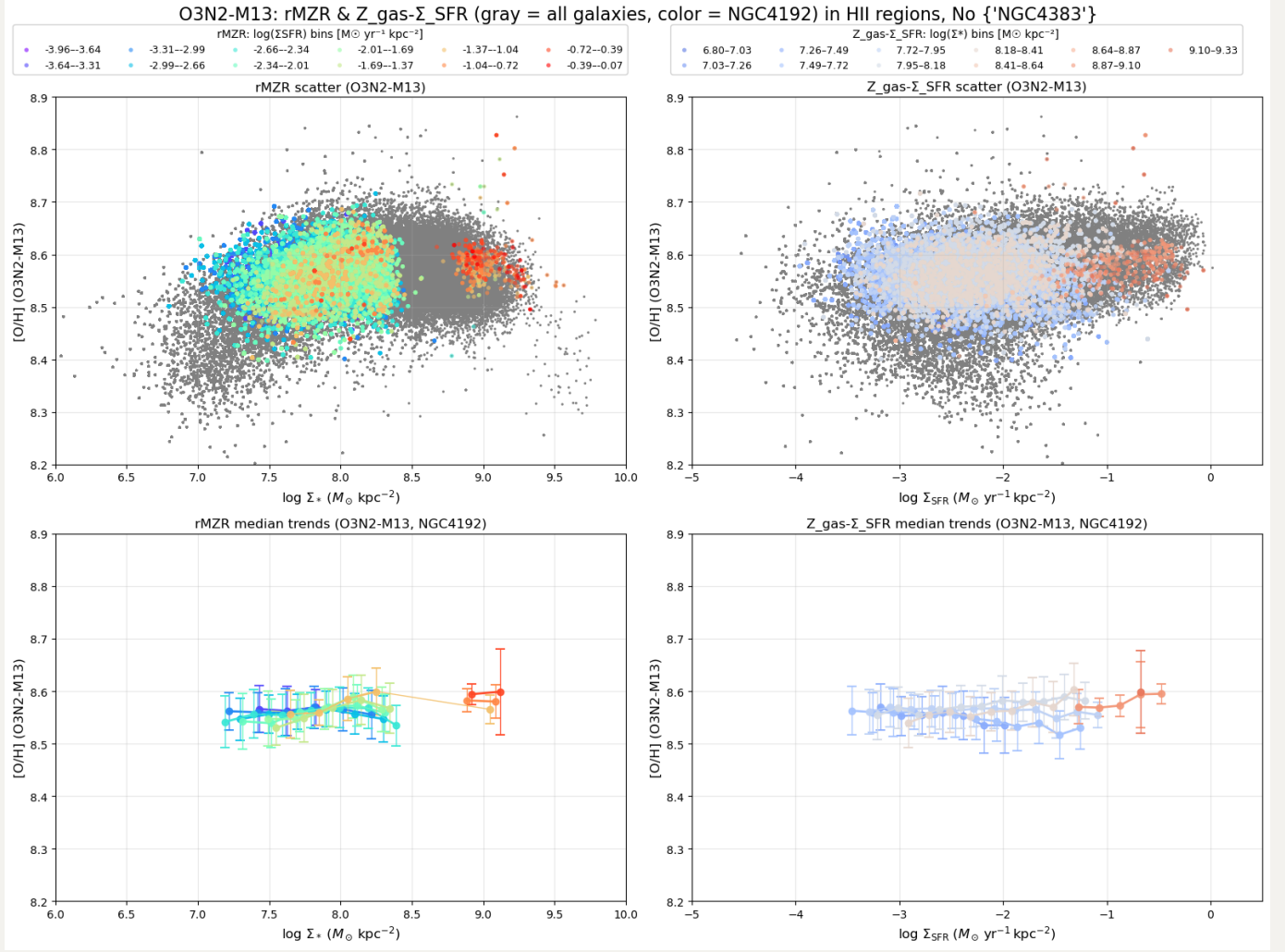


## NGC4396: Bin-Based Offset Maps with Mass Regions



## NGC4192

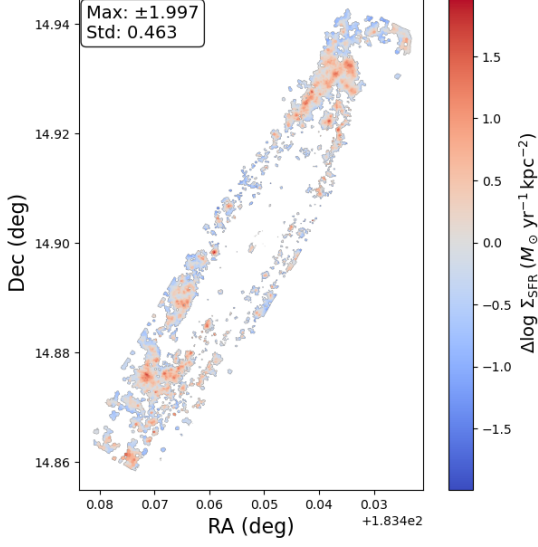
NGC4192 shows an ambiguous correlation between  $Z_{\text{gas}}$  and  $\Sigma_{\text{SFR}}$ ?



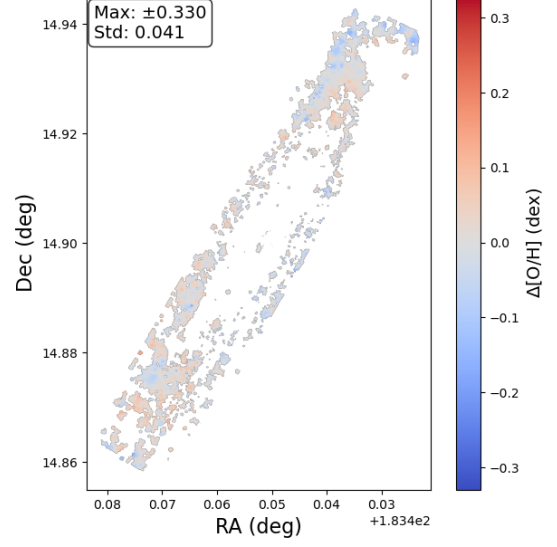
However, it shows a bit clearer inversion point at  $\Sigma_{*} \sim 7.7$ . So like the red regions ( $\Sigma_{*} \gtrsim 7.7$ ) in  $\Delta \log(\Sigma_{*})$  map show a positive correlation (redder in  $\Delta \log(\Sigma_{\text{SFR}})$  also shows redder in  $Z_{\text{gas}}$ ); while the blue regions ( $\Sigma_{*} \lesssim 7.7$ ) in  $\Delta \log(\Sigma_{*})$  map show a positive correlation (redder in  $\Delta \log(\Sigma_{\text{SFR}})$  but shows bluer in  $Z_{\text{gas}}$ ).

## NGC4192: Bin-Based Offset Maps with Mass Regions

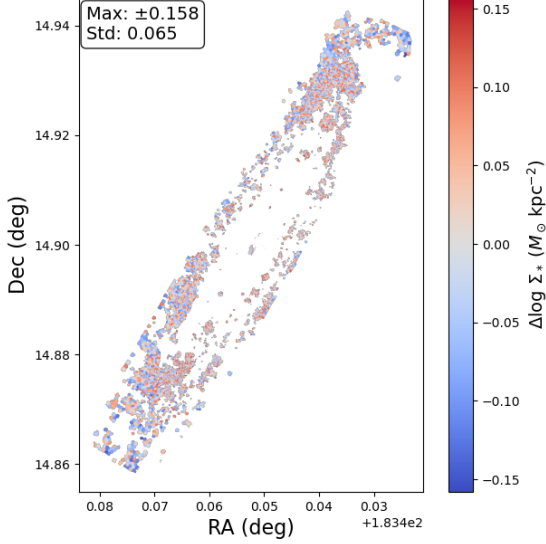
NGC4192:  $\Delta \log \Sigma_{\text{SFR}}$  (offset from bin average)



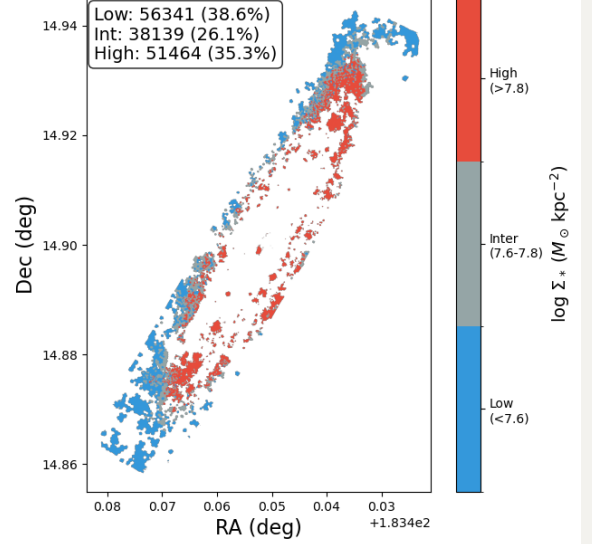
NGC4192:  $\Delta [\text{O}/\text{H}]_{\text{O3N2} - \text{M13}}$  (offset from bin average)



NGC4192:  $\Delta \log \Sigma_{*}$  (offset from bin average)



NGC4192:  $\log \Sigma_{*}$  Mass Regions

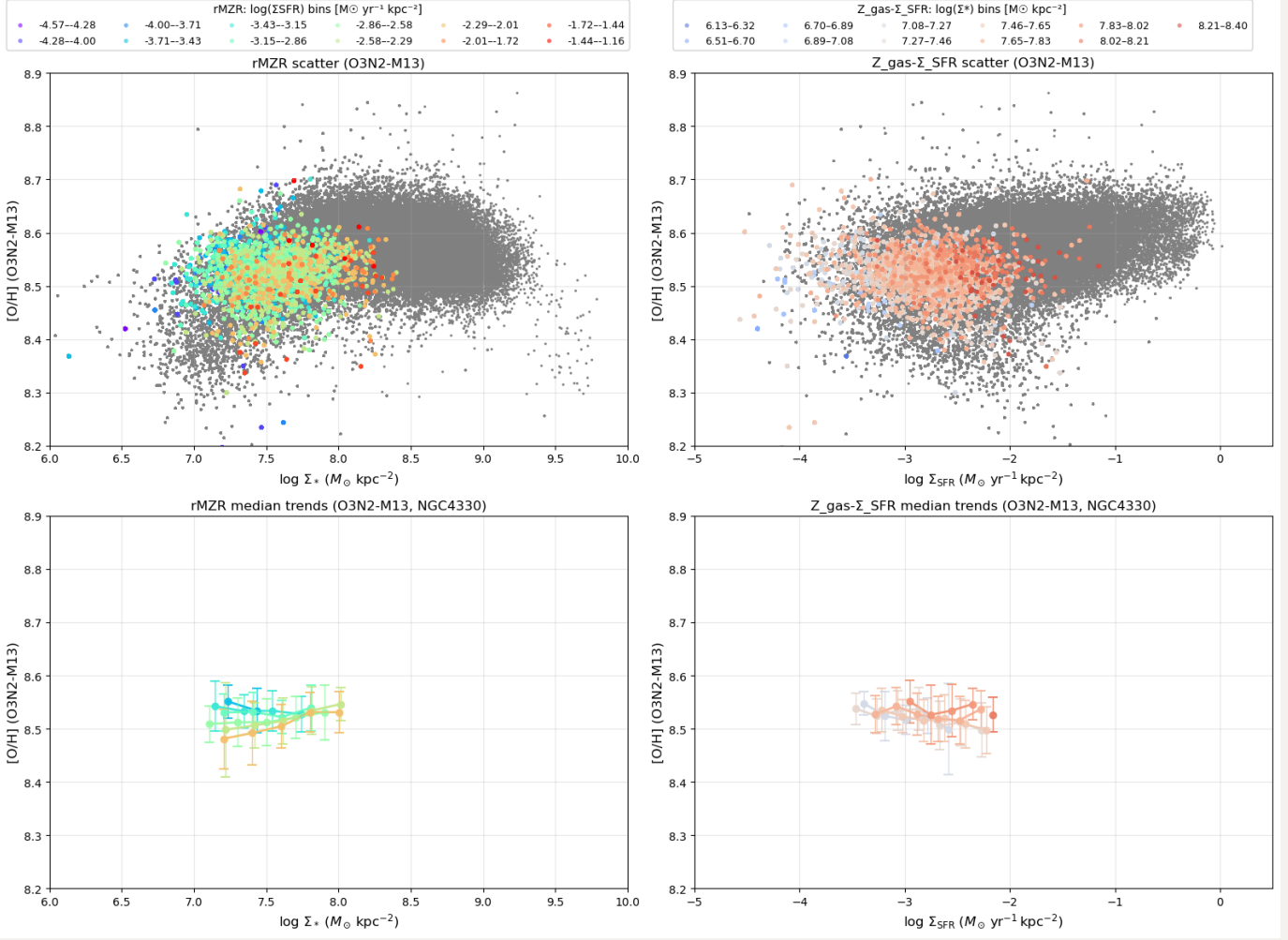


## NGC4330

NGC4330 shows a negative correlation between  $Z_{\text{gas}}$  and  $\Sigma_{\text{SFR}}$ , though not that clear.



# O3N2-M13: rMZR & Z<sub>gas</sub>-Σ<sub>SFR</sub> (gray = all galaxies, color = NGC4330) in HII regions, No {'NGC4383'}





## NGC4330: Bin-Based Offset Maps with Mass Regions

